

Iterative Progressive Retrieval Attention: Bounded Associative Closure over Latent Gist Memory

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Abstract

One-shot semantic retrieval finds memory nearest to the current query. It can miss a second piece of evidence that is weakly related to the query but strongly related to the first piece. We study iterative Progressive Retrieval Attention (PRA), which treats cached routing gists as an implicit graph: the query retrieves a bounded frontier, retrieved gists query the same layer-local index, and only the final closure is mapped to full native key/value (K/V) state. The transformer is not rerun between retrieval rounds.

A parameter-free implementation enforces depth, branch, beam, and unique-parent budgets and emits a retrieval graph before K/V materialization. Controlled two- and three-hop graphs validate chunk closure exactly, but inherited natural-data results are mixed. We therefore test two specific failure modes without retraining. First, the learned router is asymmetric: $z_q = W_q h_q$ and $z_m = W_m h_m$. Replacing the incorrect hop-2 frontier $W_m h_A$ by $W_q h_A$ recovers 0.075 QASPER any-evidence recall relative to historical closure at 10%, yet remains 0.013 below one-shot; HotpotQA chain completion is unchanged. Second, we encode 256-token parents once, derive eight 32-token local gists from each contextual state, traverse local nodes, and deduplicate parents before materializing native K/V. This local method solves every designed bridge control, where parent closure always fails. On natural data at a matched 20% parent budget, it modestly improves exact evidence identity and coverage but not chain completion, while using $7.6\times$ and $22.3\times$ the one-shot semantic comparisons on HotpotQA and QASPER. Direct iterative reuse of the inherited one-shot representation therefore does not yet provide a dependable associative traversal graph on natural data. Projection asymmetry is a real bug and local bridges are measurable, but native local query-key closure is the next gate; newly learned associative machinery is not yet justified.

1 Introduction

Suppose a question is directly related to memory chunk A . Chunk A , in turn, identifies an entity or relation needed to find chunk B . A one-shot router computes similarities from the question to all chunks. It can retrieve A while missing B because the question does not yet contain the intermediate relation. Increasing top- k may find B , but it spends the same extra budget on unrelated chunks. The problem is not transformer reasoning after evidence arrives. It is evidence discovery before full memory becomes active.

PRA already separates compact semantic routing state from full layer-native K/V payloads [8, 7, 9]. This separation suggests a narrow extension. After retrieving A , use A 's compact gist as another query. Continue for a bounded number of rounds, preserve the discovery graph, and materialize exact K/V only once. The resulting path $q \rightarrow A \rightarrow B$ is explicit. It does not require

another language-model pass, an intermediate text answer, or serialization through an external tool API.

This paper asks whether pretrained hidden-state gists already possess enough transitive semantic geometry for that extension to help. The question has a strict baseline: iterative closure must beat one-shot Top- B at the same final unique-chunk budget B . Otherwise, retrieval depth adds cost without adding a useful primitive.

The contributions are fivefold. First, we formulate iterative PRA as query-conditioned bounded associative closure over an implicit latent graph. Second, we implement Level-1 gist-to-gist and Level-2 aggregate-frontier traversal with deterministic budgets, cycle handling, root anchoring, transparent path scores, and a versioned graph artifact. Third, we evaluate five semantic-router seeds on held-out HotpotQA and QASPER features, together with exact two- and three-hop synthetic controls. Fourth, we identify and correct an asymmetric projection-interface error without retraining. Fifth, we separate contextual encoding, local propagation, and parent K/V materialization scales, showing exact local-bridge recovery but no dependable natural-data chain gain.

2 From Nearest Neighbors to Associative Closure

2.1 Why one-shot retrieval can stop too early

Let $G = \{g_1, \dots, g_N\}$ be compact routing gists at one transformer layer and let q be the current routing query. One-shot PRA returns

$$N_B(q) = \text{TopB}_{g \in G} s(q, g), \quad (1)$$

where s is cosine similarity in the learned semantic space. This operation is appropriate when every useful chunk is independently similar to q . It is incomplete when

$$s(q, g_A) \gg 0, \quad s(q, g_B) \approx 0, \quad s(g_A, g_B) \gg 0. \quad (2)$$

Here A is an entry point and B is relationally accessible only after A is known.

We interpret each chunk as a node in an implicit graph. High gist similarity supplies candidate directed edges, generated lazily when a node becomes part of the frontier. No $N \times N$ adjacency matrix is built. Iterative PRA therefore computes a bounded semantic neighborhood, not a proof and not complete reasoning.

2.2 The complete data path

Figure 1 separates traversal from payload activation. This boundary is the main systems invariant: compact gists may participate in several search rounds, while full K/V cannot enter attention until the closure stops.

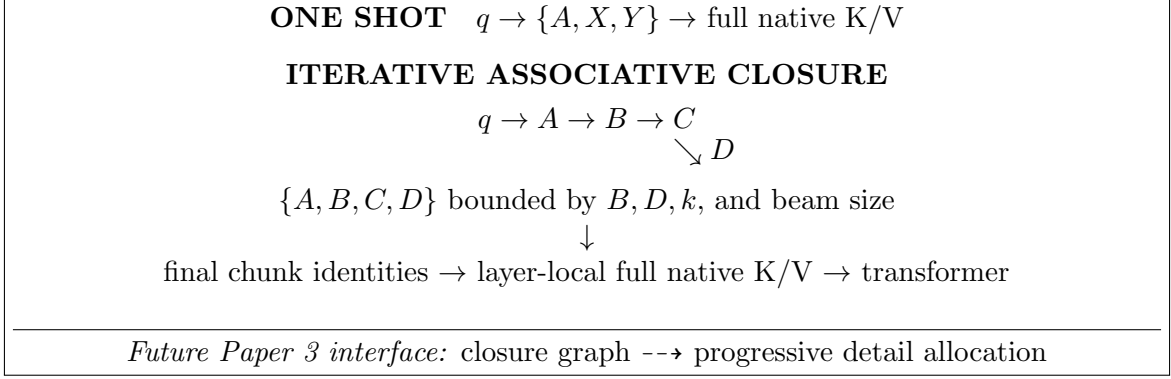


Figure 1: Iterative PRA increases search depth without transformer reruns. Gists are routing state; native K/V is the payload. The dashed handoff preserves graph signals for later work on partial materialization.

3 Method

3.1 Bounded frontier expansion

The root query initializes frontier $F_0 = \{q\}$ and the visited set is empty. At round t , each frontier vector proposes its k highest-scoring unseen chunks. Duplicate proposals are merged, the best path to each candidate is retained, and at most the remaining beam and unique-node budgets are accepted. Accepted gists form F_t . Traversal stops at fixed depth D , final budget B , no-new-node convergence, or an optional confidence threshold.

To limit semantic drift, candidate c is scored against both the root and its frontier parent:

$$e_t(c) = \alpha s(q, g_c) + (1 - \alpha) s(f_{t-1}, g_c), \quad 0 \leq \alpha \leq 1. \quad (3)$$

At $\alpha = 0$, traversal follows local gist edges. At $\alpha = 1$, every round reproduces root ranking after visited-node removal. Intermediate values preserve the original information need while allowing a retrieved node to redirect search.

The primary path score multiplies affinities $a_i = (e_i + 1)/2$ along a path:

$$P(c_t) = \prod_{i=1}^t a_i. \quad (4)$$

We also evaluate log-sum, final-edge, minimum-edge, mean-edge, and direct-root scores. These choices are deliberately transparent. A learned controller would confound the first question: whether useful graph structure is already present.

Listing 1: Level-1 bounded associative closure

```

frontier = {root}; selected = {}; graph = empty
for hop in 1..D:
    proposals = union topk_unseen(parent, G, k) for parent in frontier
    merge duplicate targets; retain parent edges and best path
    accepted = top(proposals, min(beam, B - len(selected)))
    add accepted nodes and edges to graph
    frontier = winning_gists(accepted)
    stop if frontier is empty or len(selected) == B
materialize_full_native_kv(selected) # only after closure

```

3.2 Level-2 frontier updates

Direct Level-1 traversal uses each winning gist as the next query. We compare three parameter-free updates. A residual update keeps parent state,

$$r_{t+1} = \text{normalize}(r_t + \beta g_t), \quad (5)$$

while mean and path-weighted mean collapse accepted branches to one aggregate vector and add the root. These updates test whether mild state accumulation reduces drift. They do not rerun a transformer and introduce no learned parameters.

3.3 Asymmetric projection contract

Paper 2’s frozen router does not embed questions and memories with one shared map. It computes

$$z_q = W_q h_q, \quad z_m = W_m h_m, \quad (6)$$

and scores $s(q, m) = \cos(z_q, z_m)$. The original closure stored z_m and reused it as the next frontier. Its second hop was $\cos(W_m h_A, W_m h_B)$, although the learned interface expects

$$\boxed{s(A, B) = \cos(W_q h_A, W_m h_B)}. \quad (7)$$

Equal tensor width does not make the two projections semantically interchangeable. The corrected cache derives and stores an aligned query-projected companion gist when it builds the memory index. No parameter changes and no retraining are involved. Root ranking remains $W_q h_q \rightarrow W_m h_B$; only hop 2 and later use $W_q h_A$.

3.4 Contextual parents and local propagation

Chunk means can hide a narrow relational bridge. We therefore distinguish three scales:

$$\boxed{\text{encoding granularity} \neq \text{propagation granularity} \neq \text{materialization granularity}}. \quad (8)$$

The fixed baseline encodes a 256-token parent once through Qwen. It divides that already contextualized hidden-state tensor into eight contiguous 32-token windows and mean-pools one local gist per window. Local windows are never independently re-encoded. A node has identity (parent_id, local_id) and carries its source span and both W_q and W_m projections. Traversal is

$$q \rightarrow g_{A,i} \rightarrow g_{B,j}, \quad s(A_i, B_j) = \cos(W_q h_{A,i}, W_m h_{B,j}). \quad (9)$$

Candidate local nodes are aggregated by parent, previously selected parents are removed, and the best local activation proposes each new parent at most once. Final selection materializes whole 256-token parent K/V under the same parent-count budget as one-shot and parent closure. The experiment records local nodes explored, cross-parent transitions, local and parent entropy, and

$$\Delta_{\text{bridge}} = s(g_{A,i}, g_{B,j}) - s(g_A, g_B). \quad (10)$$

3.5 Budgets, cycles, and layer locality

Four controls bound expansion: depth D , neighbors per frontier node k , per-round beam size, and maximum unique chunks B . A visited set prevents $A \rightarrow B \rightarrow A$ from consuming budget twice. Alternate parents remain graph edges even when one path wins pruning. Every index is layer-local. A query, gist, and native payload from different layers are never treated as interchangeable merely because their dimensions match.

If C is the number of chunks and $|F_t|$ the frontier size, Level-1 scoring performs $C \sum_t |F_t|$ gist comparisons. Full K/V transfer remains proportional to the final selected payload, exactly as in matched one-shot PRA. The method trades more compact-index work for a chance to find a better closure without increasing active K/V.

4 Experimental Design

4.1 Frozen semantic geometry

We reuse Paper 2’s frozen Qwen3-0.6B feature artifact and its attention-input hidden-state representation [5, 9]. The held-out split contains 16 HotpotQA and 16 QASPER examples, 32-token chunks, exact evidence-overlap masks, and 1,024-dimensional root and memory features [11, 2]. Five asymmetric linear projections, each with 262,144 parameters and seeds 11, 23, 37, 53, and 71, map query and memory features to 128 dimensions. They were trained on the disjoint Paper 2 train split with exhaustive negatives. No router is trained or selected on iterative results.

The primary setting uses direct frontier gists, multiplicative path score, $\alpha = 0.25$, and $D \in \{2, 3, 4\}$. Final budgets are 5, 10, 20, and 30% of each example’s candidate chunks. One-shot always receives the identical final B . We separately sweep $\alpha \in \{0, .25, .5, .75, 1\}$, all Level-2 updates, and all six path scores at $D = 2$, 10%.

4.2 What the evidence metrics mean

Any-evidence recall asks whether one annotated chunk is selected. Exact all-evidence recall asks whether every evidence-overlapping 32-token chunk is selected. Because one supporting sentence can cross a chunk boundary, we additionally group contiguous positive chunks and define chain completion as selecting at least one chunk from every group. The artifact reports budget feasibility for both definitions. Hotpot groups are ordered by source position; they are not claimed to encode causal hop order. QASPER usually has one contiguous evidence group and is therefore a direct-retrieval control.

The synthetic controls provide exact causal chains. Their root is similar to node A ; each next node is similar to its predecessor but is outranked from the root by distractors. At matched $B = 2$ or 3, one-shot can afford the entry node but not the full chain. We run 64 examples per seed for each chain length, yielding 320 paired trials per task.

4.3 Downstream execution validation

A separate two-example CUDA smoke loads the pinned Qwen model, indexes one HotpotQA and one QASPER source, and compares no memory, one-shot, and $D = 2$ closure. One-shot and closure both request three chunks and materialize 96 native K/V tokens. This run validates execution and budget invariants; two examples cannot estimate answer quality.

5 Results

5.1 Exact latent chains validate the mechanism

Table 1 gives the cleanest mechanism test. Both methods find some evidence, but one-shot retrieves only the entry node. Iterative traversal follows the designed edges and completes every chain. The gain costs one complete index scan per hop.

Table 1: Matched-budget synthetic results over five seeds and 64 examples per seed.

Task	Method	Any	Exact chain	Coverage	Comparisons
2-hop	One-shot Top-2	1.000	0.000	0.500	10
2-hop	Iterative $D = 2$	1.000	1.000	1.000	20
3-hop	One-shot Top-3	1.000	0.000	0.333	11
3-hop	Iterative $D = 3$	1.000	1.000	1.000	33

This result establishes that the implementation can recover indirect nodes without transformer updates. It does not show that a pretrained semantic router supplies the required edges.

5.2 HotpotQA shows a small chain signal and a larger drift cost

Figure 2 and Table 2 compare natural-data routing. At 10%, one-shot any-evidence recall is 0.700 ± 0.065 (five-seed 95% interval), while $D = 2$ obtains 0.663 ± 0.118 . Yet group-level chain completion rises from 0.138 to 0.150, and reaches 0.175 at $D = 4$. The two metrics describe different behavior: closure sometimes reaches another evidence group, but also leaves an easily retrieved entry neighborhood.

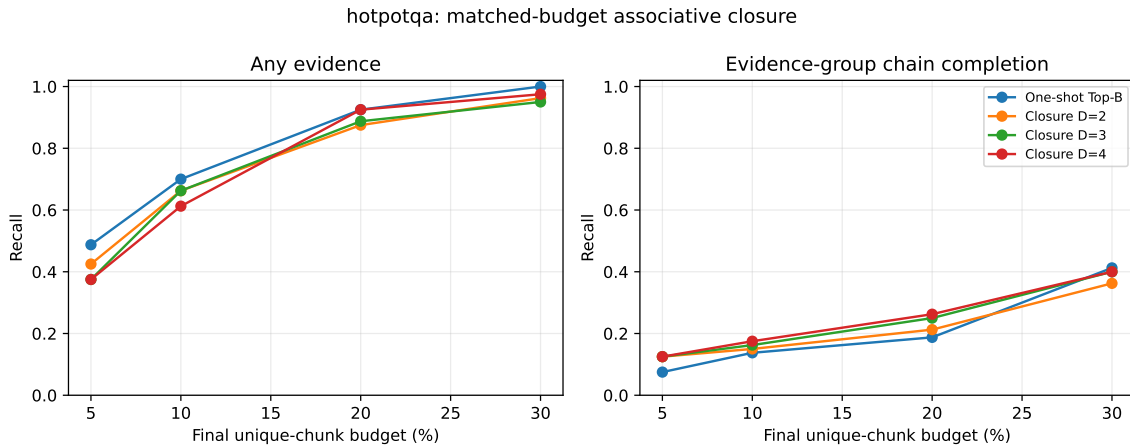


Figure 2: Five-seed HotpotQA matched-budget curves. Deeper closure modestly improves chain completion at some sparse budgets but does not improve the broader any-evidence curve.

Table 2: Primary 10% matched-budget results. Means combine 16 examples and five independently trained routers. “All” requires every positive chunk; “Chain” requires every evidence group.

Dataset	Method	Any	Chain	All	Coverage	Comparisons
HotpotQA	One-shot	.700	.138	.013	.207	50.1
HotpotQA	Closure $D = 2$	.663	.150	.000	.202	219.6
HotpotQA	Closure $D = 3$	.663	.163	.000	.205	273.0
HotpotQA	Closure $D = 4$	.613	.175	.000	.193	303.2
QASPER	One-shot	.963	.963	.213	.477	142.6
QASPER	Closure $D = 2$	.875	.875	.200	.466	1395.9
QASPER	Closure $D = 3$	.788	.788	.175	.425	1822.1
QASPER	Closure $D = 4$	.825	.825	.150	.426	2144.2

At 20%, Hotpot chain completion rises from 0.188 one-shot to 0.212, 0.250, and 0.263 at depths 2–4. At 30%, one-shot returns to the lead (0.412 versus 0.362–0.400). The effect is therefore small, budget dependent, and not accompanied by improved exact all-chunk recall. Five seeds are insufficient to claim a reliable improvement; the direction also changes with budget.

5.3 QASPER behaves as a direct-retrieval control

QASPER’s evidence commonly forms one contiguous group, so chain completion equals any-evidence recall in this protocol. Figure 3 shows that one-shot dominates throughout the sparse region. At 10%, its any/chain recall is 0.963 ± 0.069 versus 0.875 ± 0.095 for $D = 2$. Closure adds no useful intermediate relation when the root already addresses the target.

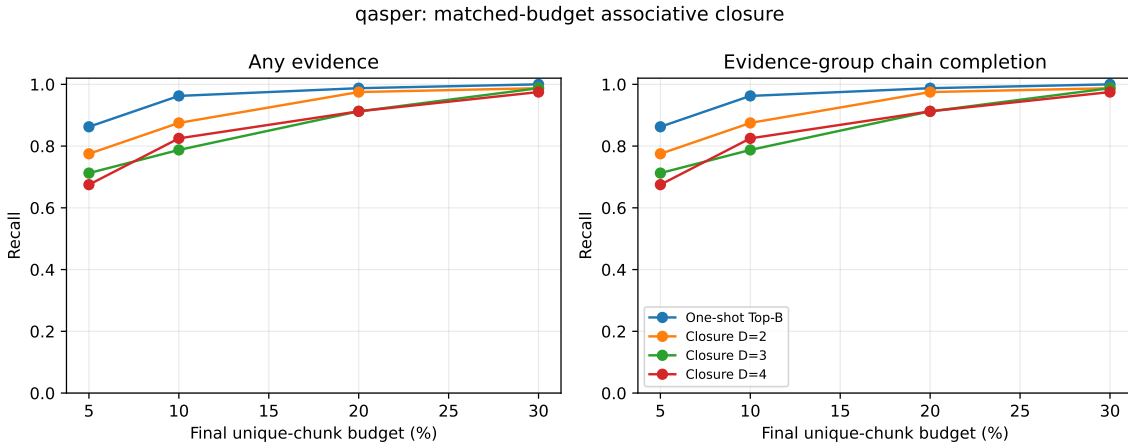


Figure 3: QASPER matched-budget curves. The mostly single-group evidence protocol favors direct one-shot ranking, as expected if recursion is unnecessary.

5.4 Root anchoring exposes semantic drift

At 10% Hotpot, pure local traversal ($\alpha = 0$) yields any recall 0.600. Increasing root weight to 0.50, 0.75, and 1.0 gives 0.700, 0.738, and 0.700. QASPER improves monotonically toward the root-only condition: 0.863, 0.875, 0.900, 0.913, and 0.963 from $\alpha = 0$ to 1. The best QASPER “iterative” setting is therefore the one that does not use frontier semantics.

Residual update is the only Level-2 variant to preserve exact all-chunk recall on Hotpot (0.013), but its any recall 0.713 does not consistently beat one-shot. Mean and weighted mean reach 0.738 any recall while exact all remains zero. Path-score variants range from 0.650 to 0.738 Hotpot any recall and do not improve chain completion consistently. No parameter-free update dominates.

5.5 Relational gap does not identify real gains

The motivating diagnostic is

$$\Delta_{rel} = \max_{A \in E \setminus \{B\}} s(g_A, g_B) - s(q, g_B). \quad (11)$$

Large Δ_{rel} should favor closure if evidence-to-evidence similarity is a useful edge. Figure 4 does not show that pattern. In Hotpot’s highest gap quartile, the paired $D = 2$ any-recall effect is -0.100 and coverage effect is $+0.017$. In QASPER they are -0.176 and -0.019 . A high pairwise cosine score is not enough to identify the correct relational transition.

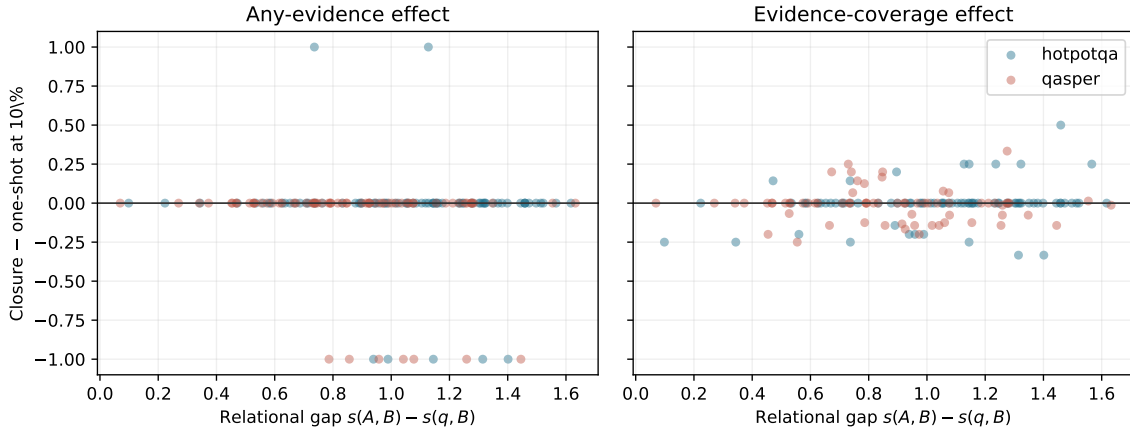


Figure 4: Paired $D = 2$ minus one-shot effects at 10%. Relational gap does not predict a stable closure advantage in the inherited semantic space.

5.6 Search cost and full-K/V execution

Figure 5 shows the compact-index cost. At 10%, $D = 2$ uses 219.6 comparisons per Hotpot example versus 50.1 one-shot ($4.4\times$), and 1395.9 versus 142.6 on QASPER ($9.8\times$). The higher QASPER multiplier follows from more candidates and larger frontiers. These are exact comparison counts; wall time depends strongly on batching and hardware.

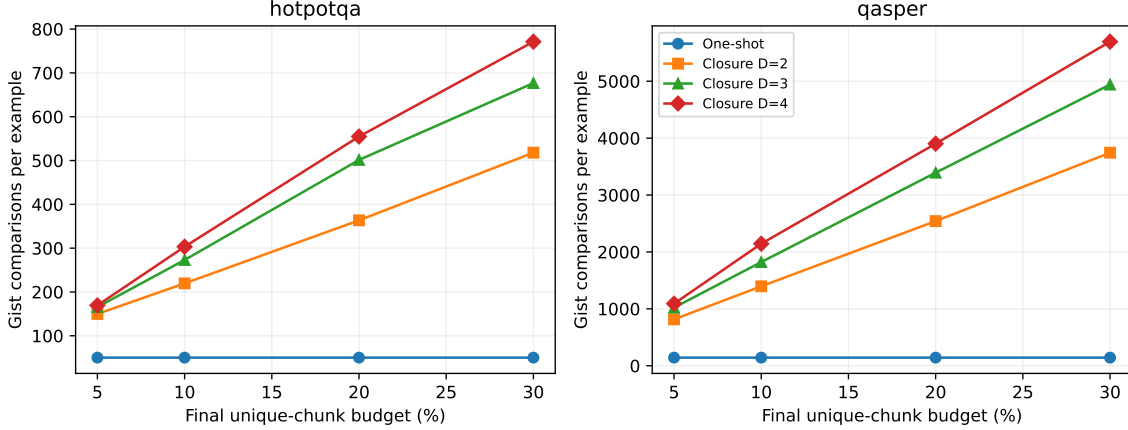


Figure 5: Compact gist comparisons. Iteration leaves final K/V budgets fixed but increases index work with frontier size and depth.

The CUDA integration smoke materializes exactly three chunks and 96 K/V tokens for both methods. Iterative routing takes 0.103/0.077 seconds on the Hotpot/QASPER examples, versus 0.107/0.073 for one-shot in this trace-oriented implementation. End-to-end times are 1.291/1.291 seconds for closure and 1.304/1.347 seconds for one-shot. The model records zero native-limit violations and a maximum 64-token native operation. These two cases validate the execution contract only.

6 Projection and Locality Gates

The preceding results freeze the original chunk-level closure as Level 1. We next change one assumption at a time. All comparisons reuse the same five checkpoints, held-out examples, depth, root anchor, path score, and final unique-parent/K/V budget. Neither gate trains a new parameter.

6.1 Gate 1: projection-correct frontiers

Table 3 compares the historical $W_m h_A$ frontier with the contract-correct $W_q h_A$ frontier at the primary 10% chunk budget. Correction materially reduces the QASPER failure: any/chain recall rises by 0.075 and coverage by 0.013 relative to historical closure. It nevertheless remains 0.013 below one-shot any/chain recall. On HotpotQA, correction raises any recall by 0.025 and restores exact identity by 0.013 relative to historical closure, but chain completion falls by 0.013 and exactly matches one-shot. Comparison counts are unchanged because this gate changes only the projection used by each activated frontier.

Table 3: Gate 1 at a matched 10% final chunk budget. Q-frontier is the corrected $W_q h_A \rightarrow W_m h_B$ traversal; M-frontier is the historical direct reuse of $W_m h_A$.

Dataset	Method	Any	Chain	Exact	Coverage	Comparisons
HotpotQA	One-shot	.700	.138	.013	.207	50.1
HotpotQA	M-frontier	.663	.150	.000	.202	219.6
HotpotQA	Q-frontier	.688	.138	.013	.201	219.6
QASPER	One-shot	.963	.963	.213	.477	142.6
QASPER	M-frontier	.875	.875	.200	.466	1395.9
QASPER	Q-frontier	.950	.950	.225	.479	1395.9

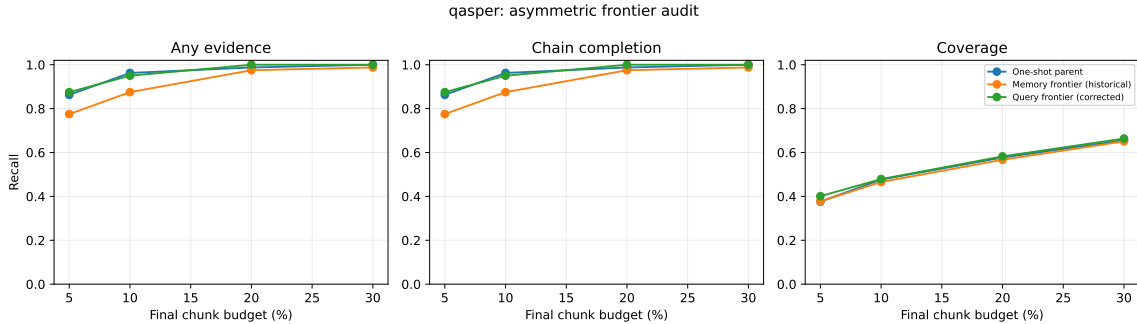


Figure 6: Gate-1 QASPER curves. Respecting the asymmetric projection contract recovers most of the historical closure loss, but one-shot remains the sparse-retrieval reference.

Projection asymmetry therefore explains a meaningful part of the original degradation. It does not explain why the corrected traversal fails to improve chain completion. That result passes the predeclared checkpoint for testing a finer propagation scale.

6.2 Gate 2: contextual local gists

Gate 2 changes the index to 256-token contextual parents with eight 32-token local gists. The one-shot and parent-closure baselines use the same parent features; local closure alone traverses subchunk states. Parent deduplication holds final materialization to the same parent count.

The designed local-bridge control separates the methods cleanly. At two selected parents, one-shot and projection-correct parent closure retrieve only the entry parent: exact identity, chain completion, and coverage are 0, 0, and 0.5 over 320 examples. Local closure retrieves both evidence groups in every case, with exact identity, chain completion, and coverage all 1.0. Its mean Δ_{bridge} is 1.0. The implementation can therefore expose a bridge erased by a parent mean without re-encoding the local window.

Natural data is more restrained (Table 4). At 20%, local closure improves Hotpot any recall, exact identity, and coverage by 0.013 each over one-shot, but chain completion falls by 0.013 and parent closure is stronger. On QASPER, local closure improves exact identity by 0.025 and coverage by 0.016 over one-shot while any/chain recall falls by 0.013. Local accepted edges are stronger than corresponding parent edges: mean Δ_{bridge} is 0.203 on Hotpot and 0.205 on QASPER. That measurable locality does not yet identify the correct cross-parent chain.

Table 4: Gate 2 at a matched 20% final parent/KV budget. Comparisons count semantic gist dot products; no native QK comparisons occur in this gate.

Dataset	Method	Any	Chain	Exact	Coverage	Gist cmp.	K/V tokens
HotpotQA	One-shot parent	.475	.138	.000	.217	6.5	405.8
HotpotQA	Parent closure	.513	.175	.000	.235	11.9	405.0
HotpotQA	Local closure	.488	.125	.013	.229	49.5	403.4
QASPER	One-shot parent	.938	.938	.738	.813	18.2	1036.0
QASPER	Parent closure	.950	.950	.738	.819	66.6	1037.3
QASPER	Local closure	.925	.925	.763	.829	405.2	1031.0

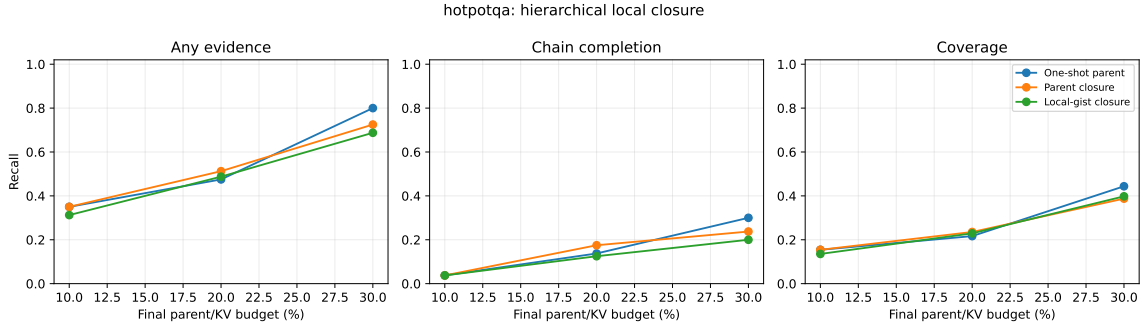


Figure 7: Gate-2 HotpotQA curves under matched final parent budgets. Local routing exposes stronger subchunk edges but does not improve evidence-group chain completion.

The cost is substantial. At 20%, local closure uses 49.5 versus 6.5 comparisons on Hotpot ($7.6\times$) and 405.2 versus 18.2 on QASPER ($22.3\times$). Materialized K/V differs by less than 1.3% within either dataset, confirming that the quality comparison is not purchased with a larger payload. The graph audit validates all 960 saved Gate-1/2 traces against schema v2 and finds no parent-budget violation.

7 Discussion

7.1 What the experiments establish

The synthetic result and the integration tests establish three mechanical facts. First, a gist can serve as a new query and recover a node that the root cannot afford at matched B . Second, deduplication, beam pruning, and hard budgets prevent exponential materialization. Third, the closure can remain entirely in compact routing state until final identities map to full native K/V.

The natural-data result establishes a different boundary. Projection-correct closure confirms that shared dimensionality did not imply shared query and memory semantics: respecting the asymmetric interface repairs much of QASPER’s loss. The local experiment then finds positive bridge deltas and better exact identity/coverage, so parent aggregation does hide some useful semantic relations. Yet those stronger local edges do not improve chain completion. Direct iterative reuse of the inherited one-shot routing representation does not yet provide a dependable associative traversal graph on natural data.

The next gate should use the transformer’s native local matching primitive rather than train a new semantic router immediately. Coarse semantic routing can narrow candidate parents, after

which pre-RoPE or otherwise position-neutral $Q_{A,i}^{(h)}K_{B,j}^{(h)\top}$ scores test whether the pretrained attention geometry carries better evidence-to-evidence transitions. Final parent and K/V budgets must remain fixed, and native comparison counts must be reported separately. Only if projection-correct local semantic and native QK closure both fail is an explicitly supervised associative head justified.

7.2 Relation to retrieval, agents, and recursive models

Memory Networks repeatedly read differentiable memory to support multi-hop tasks, while RAG, RETRO, and interleaved retrieval systems bring external evidence into generation [4, 1, 10]. Agentic systems can reformulate a text query after each model pass. Iterative PRA is narrower: it traverses already computed latent addresses inside one memory subsystem. It avoids repeated text serialization and full-model round trips, but it cannot replace an agent that must plan tools, verify evidence, or change the retrieval language.

Tiny Recursive Models repeatedly transform internal latent and answer state [3]. Controlled autoregressive work finds no reliable gain from the full autoregressive TRM architecture over simpler refinement baselines [6]. Iterative PRA does not apply a tiny neural network repeatedly. Its recurrence is *query* $\rightarrow$ *memory node* $\rightarrow$ *memory node*. This distinction motivates the parameter-free Level-1 test: retrieval recursion should earn its complexity before neural state recursion is introduced.

7.3 Paper 3 handoff

Papers 1–2 conservatively compress search but materialize every selected chunk in full. Iterative closure produces more structure before transfer: hop, parent edges, direct relevance, path score, confidence, branch entropy, and duplicate/overlap statistics. The versioned graph keeps these signals even when the final payload remains unchanged. Paper 3 can therefore ask a separate question: given a discovered closure, how much detail should each node receive? This paper does not test partial materialization.

8 Limitations

The held-out natural-data sample has 16 examples per dataset, repeated over five router seeds. Seeds quantify projection sensitivity but do not create 80 independent questions. Hotpot evidence groups are source-contiguous spans, not annotated causal hop labels. QASPER’s selected subset is yes/no and often one-group, so it tests direct retrieval more than multi-hop closure.

The natural-data sample remains small. Projection correction and local hierarchy remove two known confounds, but the experiment does not include native local QK closure, overlapping local windows, adaptive zoom rules, evidence-edge training, or a full agentic baseline. The local bridge control is deliberately geometric and does not approximate natural language. Downstream F1/EM is measured only in a two-example execution smoke and is not used as evidence of quality. Finally, the trace-oriented implementation launches many small operations; production latency requires batched frontiers, batched examples, and a persistent index service.

9 Conclusion

Bounded associative closure is exact when latent memory contains the right edges. Correcting the asymmetric frontier is necessary and materially reduces inherited degradation. Routing at a finer local scale can recover bridges erased by parent means and modestly improve exact evidence identity

and coverage. Neither correction, however, produces a dependable natural-data chain advantage at matched materialization budget, and local traversal substantially increases search work. One-shot PRA therefore remains the default efficiency/quality reference.

The scientific lesson is narrower than “semantic geometry has no associative edges.” Direct iterative reuse of a one-shot representation is sensitive to projection interface and propagation scale. Once both are corrected, local semantic affinity is still insufficient to select reliable cross-parent chains. The next experiment is hybrid semantic narrowing plus native local QK closure. Learned associative geometry remains a later gate, not the automatic response to a mixed result.

A Implementation Map

The implementation is designed so another system can reproduce the mechanism without adopting the entire PRA-HF package. File and line references below refer to the accompanying source revision.

Layer-local packed index. `src/prahf/iterative.py:69--134` packs variable gist sets as `[chunks,max_gists,routing_width]` plus a validity mask. Records retain chunk identities and payload handles, but packing copies no native K/V.

Asymmetric companions and contextual parents. Lines 387–438 of `injection.py` project each contextual local gist with both W_m and W_q and projects the occupancy-weighted parent hidden-state mean separately. The full path is `src/pratorch/hf/injection.py`. The cache stores compact companion vectors, not another native K/V payload.

Hierarchical index and parent deduplication. Lines 137–254 of `iterative.py` define aligned parent/local tensors and construct them from contextual segment gists. Lines 641–787 perform root-to-parent/local and local-to-local expansion, aggregates candidates by parent, enforces the unique-parent budget, and maps only final parents to payload handles.

Graph contract. Lines 269–337 of `iterative.py` define node, edge, graph, and result records. The current JSON Schema is `docs/papers/shared/schemas/retrieval_graph_v2.schema.json`. Nodes retain parent identity, local span, resolution, representation and projection types, hop, path score, and materialization state. Edges retain typed endpoints, score, threshold, and acceptance.

Exact scoring and deterministic top-k. `iterative.py:376--397` normalizes frontier rows and computes `einsum("fd,cgd->fcg")` before max reduction across gists. A stable index tie-break is applied before `torch.topk`, producing deterministic identities on CPU and CUDA.

Traversal. `iterative.py:399--614` performs root scoring, frontier expansion, anchoring, deduplication, path pruning, Level-2 updates, stopping, and cost accounting. The code never accesses `chunk.token_kv`. Conversion at lines 617–638 returns selected payload handles only after the graph is final.

Host-model integration. `src/prahf/model.py:374--439` captures one root query with memory disabled, dispatches one-shot, chunk iterative, or opt-in `local_iterative` routing, maps final parent identities to consumption layers, enables fixed memory, and only then marks graph nodes materialized. Generation thereafter uses the unchanged Paper 2 native-K/V path.

Reproduction. The experiment directory is `experiments/paper2_5_iterative_pra/`. Its Gate-1 runner isolates projection correction. The precompute runner captures one 256-token contextual parent pass and derives 32-token local means. The Gate-2 runner performs the matched parent-budget comparison and local-bridge control. Exact script names and commands are listed in that directory’s `README.md`; raw rows, plots, and graph JSONL are under the corresponding `shared/results/paper2_5_iterative_pra/` result tree.

B Validation Gates

The test suite checks $D = 1$ equivalence to Top- B , deterministic batches, exact synthetic two-hop recovery, deduplication and cycles, depth/beam/unique budgets, all path and frontier modes, zero limits, layer locality, projection-contract enforcement, exact local-bridge recovery, parent deduplication, CPU/CUDA identity parity, and the no-materialization-before-closure boundary. The focused iterative/HF/multi-gist suite contains 94 passing tests; 960 saved Gate-1/2 graphs validate against schema v2. Contextual local-feature capture records zero native operation-limit violations.

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